

Ajay Venkatesh

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EDUCATION

New York University	Master of Science in Computer Engineering (GPA - 3.9/4)	Graduation - May 2026
Visvesvaraya Technological University	Bachelors in ECE (GPA - 8.9/10)	Graduation - Aug 2021

EXPERIENCE

New York University, NYC, USA	Aug 2024 - Present
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Data Engineer/Analyst (On-Campus)

- Engineered an AI-driven Remote Patient Monitoring System, funded by the National Science Foundation, using React Native and TypeScript for real-time health tracking—focused on generating key insurance-related health metrics.
- Performed an ETL process to ingest unstructured NoSQL JSON patient data (10,000+ users) into a Postgres database via GCP Python Cloud Functions. Built interactive dashboards in AWS QuickSight and Q to visualize insights critical for insurance risk evaluation.
- Developed and deployed a secure React web app with an embedded QuickSight dashboard. Integrated Cognito, Amplify, and Lambda functions for role-based access control, supporting data-driven decision-making for healthcare insurers.

Price Water House Coopers, (PwC)

Senior Technology Consultant

Jul 2023 - Aug 2024

- Developed a scalable data architecture and pipeline roadmap to integrate and centralize diverse data sources for the Microsoft Dynamics Health Application, enabling enhanced analytics and decision-making capabilities.
- Designed and deployed a Deep Learning OCR model on Azure ML to automate data extraction from Images and integrate workflows via APIs, reducing manual data entry time and improving accuracy by 75%.
- Streamlined and transformed Postgres and SQL server entities for faster data preprocessing, ensuring high-quality, structured data for machine learning models, improving training efficiency and model accuracy by 30%.
- Mentored a team of 5 interns in building and optimizing data pipelines, conducting comprehensive code reviews to reduce errors by 25% and improving deployment cycles in an agile environment.
- Implemented Azure CI/CD pipelines to automate deployment workflows, achieving a 90% reduction in deployment time while ensuring consistent data pipeline operations across higher environments.

Associate Technology Consultant

Aug 2021 - Jun 2023

- Developed automated data pipelines to integrate data from 5+ sources into Dataverse, enabling real-time data synchronization and reducing data processing time by 30%.
- Collaborated with the clients to produce Power BI dashboards to monitor and analyze, data, established multi-level data filtering and security rules to access data, and improved data visualization and analysis efficiency.
- Improved an automation flow used to clean, map, and import 100K rows of data from an Excel file to the Database from 5 hours to 1 hour, resulting in an 80% increase in efficiency.
- Designed and implemented a MySQL database to securely store sensitive client data, reducing unauthorized access risks by 40% through role-based access controls and encryption.
- Established a robust version control system using git in pipeline projects at PwC; streamlined workflows resulted in a reduction of integration issues encountered during deployments by over 25%.

PROJECTS

Executive Performance Analytics Dashboard

- Architected a data pipeline using Power Automate to process and transform quarterly, monthly, and yearly reports, ensuring seamless data ingestion and transformation for analysis in Power BI.
- Generated interactive dashboards in Power BI and Canvas Apps, enabling directors to analyze key business metrics, track trends, and gain actionable insights from structured reports, enhancing reporting efficiency by 30%.

Hospital Management System

- Designed a React application to manage Appointments, Patient and Doctor records, Lab Reports and Medical History.
- Implemented MySQL data consistency measures via triggers and stored procedures, reducing anomalies by 50% and enhancing performance by 40%.

SKILLS

Programming languages: Python(NumPy, Pandas, Polars, Dash) , Java, JavaScript, SQL, C#.

Database: RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON).

Cloud: AWS (S3, Amplify, Lambda Function, API Gateway, Cognito, QuickSight, Q, SageMaker) **Azure** (AI services, Logic Apps, Databricks, Function Apps, Storage blobs)

Tools: Git, Azure DevOps, Hadoop, Power Automate, Power BI, Tableau. Apache, Spark, Adobe.

Methodology: Agile, Waterfall.

AI/ML: Machine Learning, Deep Learning, Big Data, ML-Ops, Principal component analysis.

CERTIFICATIONS

Power BI Data Analyst Associate	Azure Data Fundamentals	Azure Data Scientist Associate
AWS Cloud Practitioner	Power Platform Solution Architect Expert	Power Platform Developer Associate

TRAININGS

AI-102: Designing and implementing a Microsoft Azure AI Solution (PWC)	Apr 2022 - May 2024
Microsoft Power Apps, Power BI training and Azure (PWC)	Jan 2022 - Apr 2022

PUBLICATIONS

Supervised Machine Learning System Based Segmentation and Classification of Stroke Using Deep Learning Techniques ([IEEE](#))
An Aquila-optimized SVM classifier for Diabetes prediction ([IEEE](#))